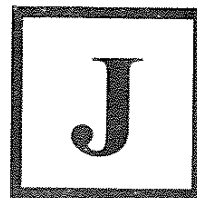


THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Bhaskara Contest

(NMTC- at **JUNIOR LEVEL** IX & X Standards)

Saturday, 31st August 2013.



Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
- 2) Diagrams are only visual aids; they are not drawn to scale.
- 3) You are free to do rough work on separate sheets.
- 4) Duration of the test: 2 p.m. to 4 p.m.- 2 hours.

PART – A

Note:

- Only one of the choices A,B,C,D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

1. Two sides of a triangle are 10cm and 5cm in length and the length of the median to the third side is $6\frac{1}{2}$ cm. The area of the triangle is $6\sqrt{x}$ cm². The value of x is

(A) 12 (B) 13 (C) 14 (D) 15

2. x and y are real numbers such that $7^x - 16y = 0$ and $4^x - 49y = 0$, then the value of $(y - x)$ is

(A) $\frac{5}{2}$ (B) $\frac{19}{5}$ (C) $\frac{4115}{2013}$ (D) $\frac{1569}{784}$

3. a, b are positive integers such that

- (i) the sum of their squares is 5,
- (ii) the sum of their cubes is C times the sum of the numbers itself.
- (iii) $S \cdot C = 28$. The number of such pairs (a, b) is

(A) 1 (B) 2 (C) 3 (D) 6

4. The number of numbers of the form $30a0b03$ where a, b are digits which are divisible by 13 is

(A) 5 (B) 6 (C) 7 (D) 0

5. The number of positive integral values of (x, y) which satisfy the equation $\sqrt{x} + \sqrt[3]{y} = 4$; $x + y = 28$ simultaneously is

(A) 1 (B) 2 (C) 0 (D) 3

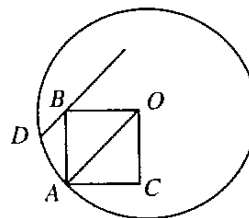
6. $ABCD$ is a rectangle. Through C a variable line is drawn so as to cut AB at X and DA produced at Y . then $BX \cdot DY$ is

- (A) twice the area of the rectangle $ABCD$.
 (B) equal to the area of the rectangle $ABCD$.
 (C) a variable quantity which lies between the area of rectangle $ABCD$ and twice the area of the rectangle $ABCD$
 (D) always a constant less than the area of rectangle $ABCD$.

7. The number of ordered triples (x, y, z) such that x, y, z are primes and $x^y + 1 = z$ is

- (A) 0 (B) 1 (C) 2 (D) infinitely many.

8. In the adjoining figure O is the centre of the circle. $ACOB$ is a square with A on the circle. Through B a line parallel to OA is drawn to cut the circle at D nearer to A . Then $\angle BOD =$

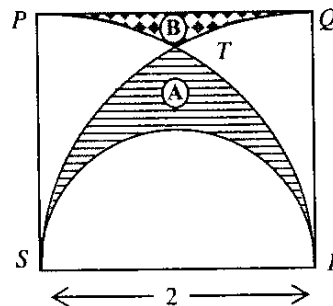


- (A) 20° (B) 18° (C) 15° (D) $22\frac{1}{2}^\circ$

9. The number of real solutions of the equation $x + \sqrt{x^2 + \sqrt{x^3 + 1}} = 1$ is

- (A) 1 (B) 2 (C) 3 (D) 0

10. In the adjoining figure $PQRS$ is a square of side 2 units. PTR and QTS are quadrants of circles of radius 2 units. With SR as diameter a semicircle is drawn. A, B denote the areas of the portions shaded. Then $(A - B) =$



- (A) $\frac{3}{2} - 4$ (B) $-\frac{1}{3}$ (C) $\frac{3}{2} - \frac{1}{4}$ (D) 4

11. a, b, c are digits of a 3-digit number such that $64a + 8b + c = 403$, then the value of $a + b + c + 2013$ is

- (A) 2024 (B) 2025 (C) 2034 (D) 2035

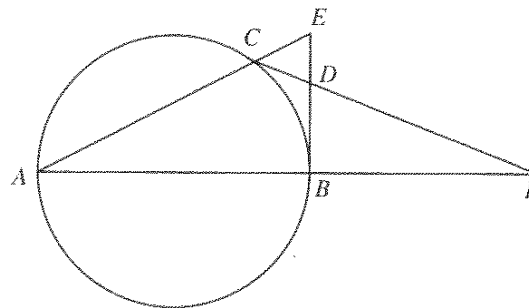
12. What is the sum of the digits of $(9999999999)^3$

- (A) 99 (B) 108 (C) 180 (D) 199

13. The number of three digit numbers which are divisible by 3 and have the additional property that the sum of their digits is 4 times their middle digit is

- (A) 7 (B) 4 (C) 11 (D) 10

14. In the adjoining figure AB is a diameter of a circle. AB is produced to P such that $BP =$ radius of the circle. PC is a tangent to the circle. The tangent at B and AC produced cut at E . Then CDE is



- (A) isosceles with $EC = ED$. (B) isosceles with $EC = CD$
(C) equilateral. (D) a scalene triangle.

15. Nine numbers are written in ascending order. The middle number is the average of the nine numbers. The average of the five largest numbers is 68 and the average of the five smallest numbers is 44. The sum of all numbers is

- (A) 560 (B) 504 (C) 112 (D) 122

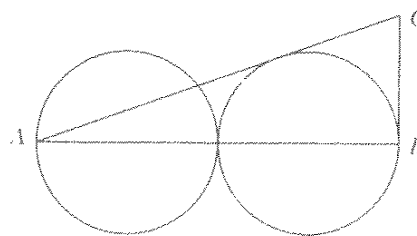
PART – B

Note:

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

16. The least value of the positive integer ' n ' such that $(n+20) + (n+21) + (n+22) + \dots + (n+100)$ is a perfect square is _____.

17. In the adjoining figure two equal circles of radii 2 units each touch. AB is the Common diameter. The tangent at B meets the tangent from A to circle at C as shown. If $BC = K\sqrt{2}$ then the value of K is _____.



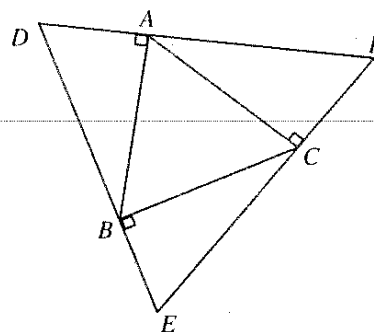
18. When the number $33333^2 + 22222$ is written as a single decimal number, the sum of its digits is _____.

19. The number of three digit numbers such that the product of their digits is a prime number is _____.

20. The number of real values (x, y) for which $2^{x+1} + 3^y = 3^{y+2} - 2^x$ is _____.

21. In the adjoining figure, ABC is equilateral. AD, BE and CF are respectively perpendicular to AB, BC and AC . Then

$$\frac{\text{Area of } DEF}{\text{Area of } ABC} = \underline{\hspace{2cm}}$$



22. If $f(x) = ax + b$ and $f(f(f(x))) = 27x + 26$ then $(a + b) = \underline{\hspace{2cm}}$.

23. The eight digits 6, 5, 5, 4, 4, 3, 2 and 1 are used to form two 3-digit numbers and one 2-digit number. The largest possible sum of these numbers is $\underline{\hspace{2cm}}$.

24. $a \neq 0, b \neq 0$ The number of real number pairs (a, b) which satisfy the equation $a^4 + b^4 = (a + b)^4$ is $\underline{\hspace{2cm}}$.

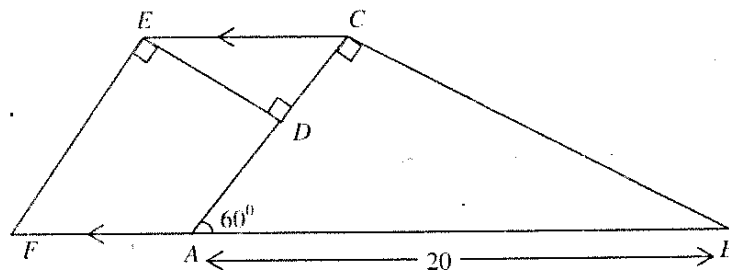
25. The number of integers greater than 2 and less than 70 that can be written as a^b (where $b \geq 1$) is $\underline{\hspace{2cm}}$.

26. $ABCD$ is a parallelogram P is a point on AD such that $\frac{AP}{AD} = \frac{1}{2013}$. Q is the point of intersection of AC and BP . Then $\frac{AQ}{AC} = \underline{\hspace{2cm}}$.

27. $ABCD$ is a square. E and F are respectively points on BC and CD such that $\angle EAF = 45^\circ$. AE and AF cut the diagonal BD at P, Q respectively. Then $\frac{\text{Area of } AEF}{\text{Area of } APQ} = \underline{\hspace{2cm}}$.

28. m, n are natural numbers. The number of pairs (m, n) for which $m^2 + n^2 - 2mn - 2013m - 2013n + 2014 = 0$ is $\underline{\hspace{2cm}}$.

29. In the adjoining figure BAC is a $30^\circ - 60^\circ - 90^\circ$ triangle. D is the midpoint of AC . The perpendicular at D to AC meets the line parallel to AB through C at E . the line through E perpendicular to DE meets BA produced at F . If $DF = 5\sqrt{x}$ then $x = \underline{\hspace{2cm}}$.



30. PR and PQ are tangents to a circle and QS is a diameter. Then $\frac{QPR}{RQS} = \underline{\hspace{2cm}}$.